

João Paulo Antunes de Souza

Full Stack Developer · Computer Engineer

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ABOUT ME

Computer Engineer and Full Stack Developer with 6+ years of experience building cloud software for precision agriculture — from React/Next.js interfaces and serverless Python APIs on AWS to the CI server that builds firmware for autonomous farm machines. I own features end to end: data model, API, user interface, deployment, monitoring and cost. 1,000+ merged pull requests across the company's products. Experienced with geospatial data, real-time IoT telemetry and AI-assisted development (Claude Code, ChatGPT).

WORK EXPERIENCE

08/2022 – Present

Full Stack Developer (Mid-level)

Zait (formerly Smart Agri / Smart Sensing) — Piracicaba, Brazil

Work across several of the company's precision agriculture products.

- Coverage path planning over field boundaries with Fields2Cover and Shapely: obstacle no-go zones, AB-line and fixed-angle modes, sampling routes; route import/export (GeoJSON, QGIS, interactive HTML map) and large uploads through S3 presigned POST.
- Missions and multi-machine operations: field-bound mission templates, tasks shared by two or more machines, refuelling estimates and live telemetry in the web UI (Next.js, React, TypeScript, Leaflet).
- Authentication and role-based access control on Amazon Cognito: user lifecycle, permission groups per field and machine, user banning enforced in server-side JWT validation.
- Machine-failure alerts over the WhatsApp Business Cloud API (Meta): speed/pressure quality alerts, per-machine recipients, deduplication, and an interactive menu to pause and resume machines remotely with confirmation.
- Designed and built a self-hosted CI server (Python, systemd, Docker) that builds and publishes firmware for 13 embedded projects (ESP-IDF, PlatformIO, STM32) with reproducible builds and byte-level parity checks, replacing AWS CodeBuild; plus a backend deploy lane with rollback and Discord notifications.
- AWS cost reduction driven by measurement (Cost Explorer, CloudWatch): Lambda right-sizing, telemetry split into latest-state and history with TTL retention, key lookups instead of scans, S3 lifecycle rules, IoT Core message cost analysis.
- Field and operation reports rendered in Lambda (Matplotlib, Playwright), GeoTIFF layers, report delivery over WhatsApp, daily AWS cost dashboard, permission management and expiring shareable links.
- Grower > farm > field hierarchy modelled with DynamoDB single-table design; infrastructure as code with AWS CDK (Python).

12/2021 – 08/2022

Full Stack Developer (Junior)

Zait (formerly Smart Agri) — Piracicaba, Brazil (hybrid)

- Map-based web interfaces in React and Leaflet: GeoJSON layers, lasso selection and drone orthomosaic layers.
- REST APIs in Python (FastAPI) on AWS Lambda and API Gateway with DynamoDB, S3 and Cognito; unit tests; PostgreSQL, MySQL and DynamoDB data modelling.

12/2020 – 12/2021

IT Intern

Zait (formerly Smart Agri) — Piracicaba, Brazil

- REST APIs in Python (FastAPI) and React interfaces; Docker and AWS serverless services (Lambda, API Gateway, DynamoDB, S3).
- Database modelling and unit testing.

06/2019 – 09/2020

Junior Systems Analyst

Farm Solutions — Piracicaba, Brazil

- Web development (HTML, CSS, JavaScript) with Git version control.

EDUCATION AND TRAINING

01/2018 – 12/2023

Bachelor's Degree in Computer Engineering

Federal Institute of Education, Science and Technology of São Paulo (IFSP) — Brazil

- Competitive programming training group.
- Windows application development group (C#, MySQL).

2015 – 2017

Technical Course in Electronics

Etec Prof. Armando Bayeux da Silva — Brazil

LANGUAGE SKILLS

Mother tongue: Portuguese

	Listening	Reading	Spoken production	Spoken interaction	Writing
English	B1	C1	B1	B1	B2

Levels: A1/A2 basic · B1/B2 independent · C1/C2 proficient (Common European Framework of Reference)

DIGITAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, C#

Front end: React, Next.js, React Native, Leaflet, HTML, CSS

Back end & cloud: FastAPI, REST APIs, AWS (Lambda, API Gateway, DynamoDB, Cognito, S3, IoT Core/MQTT, CloudWatch, EventBridge, Secrets Manager, ECR, CodeBuild, Cost Explorer), AWS CDK, Docker

Data & GIS: DynamoDB single-table design, PostgreSQL, MySQL, GeoJSON, Shapely, Fields2Cover, GeoTIFF, QGIS

DevOps & quality: Git, GitHub, CI/CD, pytest, pre-commit, Linux, systemd

Embedded: ESP-IDF, PlatformIO, STM32 (build tooling), MicroPython, ESP32, Raspberry Pi

AI tools: AI-assisted development with Claude Code and ChatGPT

Integrations: WhatsApp Business Cloud API, Discord webhooks

PROJECTS

Oh My Map: Web map application in JavaScript — oh-my-map.vercel.app